Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.15	20.22	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.02	+0.3%
power (W), mean	629	641	+1.9%
power (W), peak	650	684	+5.3%
energy (Wh)	211	216	+2.2%
energy, parked workers still on at idle power (Wh)	211	216	+2.2%
CPU used (cores), mean	0.898	1.20	+33.9%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.037	0.050	+33.9%
energy per core-hour (Wh)	701	533	-23.9%
node-hours per core-hour	6.68	4.99	-25.3%
pending pods, pod-minutes	0.517	0.267	-48.4%
pending pods, peak	1.00	1.00	+0.0%
HPA replicas, mean	9.17	9.34	+1.8%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.517	0.800	+54.8%
requests timed	1,045	1,100	+5.3%
time over the response line (% of samples)	0.287	0.000	-100.0%
response time (ms), mean	153	94.50	-38.3%
response time (ms), median	129	80.80	-37.5%
response time (ms), 95th percentile	302	153	-49.3%
response time (ms), 99th percentile	444	201	-54.7%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.2137	5.7584	-20.2%	[-3.1192, +0.1965]	EVEN, no significant difference
kwh_per_core_hour	0.7536	0.6093	-19.1%	[-0.3037, +0.0184]	EVEN, no significant difference
utilisation	0.0370	0.0492	+33.2%	[+0.0001, +0.0257]	GAIN
pending_pod_minutes_per_hour	1.4877	0.7619	-48.8%	[-2.9754, +1.5238]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	1.4762	2.2857	+54.8%	[-2.2381, +4.5714]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.358 0.392 0.421 0.446 0.468 0.488 0.505 0.519 0.533 0.546 0.557 0.562 0.568 0.572 0.577 0.581 0.586 0.589 0.59 0.591
- nodes decided per minute: 6 6 5 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	617	621	5.00	6.00	0.000	0.000	0.559	0.640

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	620	623	6.00	6.75	0.000	0.000	0.651	0.742
2	6.00	6.00	618	620	7.00	7.00	0.000	0.000	0.610	0.662
3	6.00	6.00	626	635	7.00	7.50	0.000	0.250	0.798	1.06
4	6.00	6.00	633	646	9.00	10.00	0.250	0.000	1.01	1.34
5	6.00	6.00	636	648	10.00	10.00	0.000	0.000	1.08	1.39
6	6.00	6.00	633	647	10.00	10.00	0.000	0.000	0.984	1.35
7	6.00	6.00	645	675	10.00	10.00	0.000	0.000	1.32	2.10
8	6.00	6.00	647	675	10.00	10.00	0.000	0.000	1.36	2.13
9	6.00	6.00	646	672	10.00	10.00	0.000	0.000	1.35	2.02
10	6.00	6.00	646	656	10.00	10.00	0.000	0.000	1.34	1.61
11	6.00	6.00	616	626	10.00	10.00	0.000	0.000	0.538	0.784
12	6.00	6.00	618	621	10.00	10.00	0.000	0.000	0.598	0.680
13	6.00	6.00	623	632	10.00	10.00	0.250	0.000	0.747	0.965
14	6.00	6.00	633	650	10.00	10.00	0.000	0.000	1.00	1.46
15	6.00	6.00	635	653	10.00	10.00	0.000	0.000	1.04	1.44
16	6.00	6.00	632	648	10.00	10.00	0.000	0.000	0.995	1.37
17	6.00	6.00	629	637	10.00	10.00	0.000	0.000	0.891	1.11
18	6.00	6.00	618	623	10.00	10.00	0.000	0.000	0.608	0.715
19	6.00	6.00	620	622	10.00	10.00	0.000	0.000	0.648	0.727

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.